Year 10 Higher

REVISION PACK

Number 1

Name: _____

Class: _____

Number 1 · Working with fractions (Lesson 1.1)

Worked example: adding fractions

Work out $\frac{3}{4} + \frac{5}{6}$

Give your answer as a mixed number.

Step 1	Make the bottoms the same	$\frac{9}{12} + \frac{10}{12}$
Step 2	Add the tops	$\frac{19}{12}$
Step 3	Answer	$1\frac{7}{12}$

Practice: adding fractions

1.1 Work out $\frac{4}{5} + \frac{2}{3}$. Give your answer as a mixed number.

1.2 Work out $\frac{5}{6} + \frac{3}{8}$. Give your answer as a mixed number.

1.3 Work out $\frac{3}{4} + \frac{2}{3}$. Give your answer as a mixed number.

1.4 Work out $\frac{5}{7} + \frac{1}{2}$. Give your answer as a mixed number.

1.5 Work out $\frac{7}{8} + \frac{1}{3}$. Give your answer as a mixed number.

1.6 Work out $\frac{5}{9} + \frac{5}{6}$. Give your answer as a mixed number.

Number 1 · Working with fractions (Lesson 1.1)

Worked example: multiplying mixed numbers

Work out $1\frac{2}{3} \times 2\frac{1}{4}$

Give your answer as a mixed number in its simplest form.

Step 1	Improper fractions	$\frac{5}{3} \times \frac{9}{4}$
Step 2	Tops ×, bottoms ×	$\frac{45}{12}$
Step 3	Divide top and bottom by 3	$\frac{15}{4}$
Step 4	Answer	$3\frac{3}{4}$

Practice: multiplying mixed numbers

1.7 Work out $1\frac{1}{2} \times 2\frac{2}{3}$

1.8 Work out $3\frac{1}{3} \times 1\frac{3}{5}$. Give your answer as a mixed number in its simplest form.

1.9 Work out $2\frac{1}{4} \times 1\frac{5}{6}$. Give your answer as a mixed number in its simplest form.

1.10 Work out $1\frac{3}{4} \times 2\frac{2}{5}$. Give your answer as a mixed number in its simplest form.

1.11 Work out $2\frac{2}{3} \times 1\frac{1}{8}$

1.12 Work out $3\frac{3}{4} \times 1\frac{2}{3}$. Give your answer as a mixed number in its simplest form.

Number 1 · Working with fractions (Lesson 1.1)

Worked example: dividing mixed numbers

Work out $2\frac{1}{4} \div 1\frac{1}{2}$

Step 1	Improper fractions	$\frac{9}{4} \div \frac{3}{2}$
Step 2	Flip and multiply	$\frac{9}{4} \times \frac{2}{3} = \frac{18}{12}$
Step 3	Divide top and bottom by 6	$\frac{3}{2}$
Step 4	Answer	$1\frac{1}{2}$

Practice: dividing mixed numbers

1.13 Work out $4\frac{1}{2} \div 1\frac{1}{5}$. Give your answer as a mixed number in its simplest form.

1.14 Work out $2\frac{2}{3} \div 1\frac{1}{3}$

1.15 Work out $3\frac{3}{4} \div 2\frac{1}{2}$. Give your answer as a mixed number in its simplest form.

1.16 Work out $5\frac{1}{4} \div 1\frac{3}{4}$

1.17 Work out $2\frac{2}{5} \div 1\frac{1}{5}$

1.18 Work out $4\frac{1}{6} \div 1\frac{2}{3}$. Give your answer as a mixed number in its simplest form.

Number 1 · Working with fractions (Lesson 1.1)

Worked example: subtracting with a mixed number

Work out $2\frac{1}{4} - \frac{2}{3}$

Step 1	Improper fraction	$\frac{9}{4} - \frac{2}{3}$
Step 2	Make the bottoms the same	$\frac{27}{12} - \frac{8}{12}$
Step 3	Subtract the tops	$\frac{19}{12}$
Step 4	Answer	$1\frac{7}{12}$

Practice: subtracting with a mixed number

1.19 Work out $2\frac{1}{3} - \frac{5}{6}$

1.20 Work out $1\frac{3}{5} - \frac{3}{4}$

1.21 Work out $3\frac{1}{4} - \frac{2}{3}$

1.22 Work out $2\frac{2}{5} - \frac{5}{6}$

1.23 Work out $1\frac{5}{6} - \frac{3}{4}$

1.24 Work out $2\frac{1}{2} - \frac{4}{5}$

Number 1 · Working with fractions (Lesson 1.1)

Worked example: fraction problems

Sara spends $\frac{1}{3}$ of the money on rent. Sara spends $\frac{1}{4}$ of the rest on food. What fraction of the money does Sara have left?

Step 1	The rest: 1 − ⅓	$1 - \frac{1}{3} = \frac{2}{3}$
Step 2	Food: ¼ of the rest	$\frac{1}{4} \times \frac{2}{3} = \frac{1}{6}$
Step 3	Left: rest − food	$\frac{2}{3} - \frac{1}{6} = \frac{1}{2}$
Step 4	Answer	$\frac{1}{2}$

Practice: fraction problems

1.25 Omar spends $\frac{2}{5}$ of the money on rent. Omar spends $\frac{1}{3}$ of the rest on food. What fraction of the money does Omar have left?

1.26 Huda spends $\frac{1}{4}$ of the money on clothes. Huda spends $\frac{2}{3}$ of the rest on books. What fraction of the money does Huda have left?

1.27 Yousef spends $\frac{3}{8}$ of the money on a phone. Yousef spends $\frac{1}{5}$ of the rest on games. What fraction of the money does Yousef have left?

1.28 Mona spends $\frac{1}{6}$ of the money on food. Mona spends $\frac{3}{5}$ of the rest on travel. What fraction of the money does Mona have left?

1.29 Fahad spends $\frac{3}{7}$ of the money on rent. Fahad spends $\frac{1}{2}$ of the rest on food. What fraction of the money does Fahad have left?

1.30 Laila spends $\frac{2}{9}$ of the money on books. Laila spends $\frac{3}{7}$ of the rest on clothes. What fraction of the money does Laila have left?

Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)

Worked example: order of operations

This question is about the order of operations.

(a) Work out $2 + 4 \times 3^2$

(b) Insert one pair of brackets to make this calculation correct.
$8 + 2 \times 5 - 1 = 49$

(a)	Step 1	Indices first	$3^2 = 9$
	Step 2	Then multiply	$4 \times 9 = 36$
	Step 3	Then add	$2 + 36 = 38$
(b)	Step 1	Brackets round 8 + 2	$(8 + 2) \times 5 - 1$
	Step 2	Check	$10 \times 5 - 1 = 49$

Practice: order of operations

2.1 This question is about the order of operations.

(a) Work out $7 + 2 \times 5^2$

(b) Insert one pair of brackets to make this calculation correct. $3 + 5 \times 4 - 2 = 30$

2.2 This question is about the order of operations.

(a) Work out $20 - 3 \times 2^3$

(b) Insert one pair of brackets to make this calculation correct. $2 \times 6 + 3 - 1 = 17$

2.3 This question is about the order of operations.

(a) Work out $6 + 18 \div 3^2$

(b) Insert one pair of brackets to make this calculation correct. $4 + 8 \times 2 + 1 = 28$

2.4 This question is about the order of operations.

(a) Work out $9 + 4 \times 2^3$

(b) Insert one pair of brackets to make this calculation correct. $5 \times 4 - 1 + 2 = 17$

2.5 This question is about the order of operations.

(a) Work out $3 + 6 \times 2^2$

(b) Insert one pair of brackets to make this calculation correct. $10 - 2 \times 3 + 1 = 25$

2.6 This question is about the order of operations.

(a) Work out $40 - 5 \times 3^2$

(b) Insert one pair of brackets to make this calculation correct. $6 + 2 \times 7 - 3 = 53$

Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)

Worked example: rounding

Round 562.738

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

(a)	**Step 1**	Next digit is 3: round down	562.7
(b)	**Step 1**	Next digit is 2: round down	560

Practice: rounding

2.7 Round 275.846

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

2.8 Round 1093.472

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

2.9 Round 48.0961

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

2.10 Round 736.258

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

2.11 Round 5281.63

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

2.12 Round 19.954

(a) correct to 1 decimal place. (b) correct to 2 significant figures.